From Above and Below: Assessing the Impacts of Droughts & Storm Surges on Coastal Wetland Health

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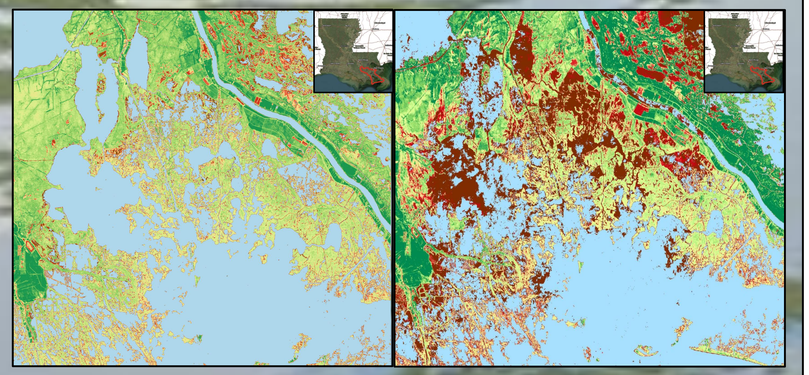
Background

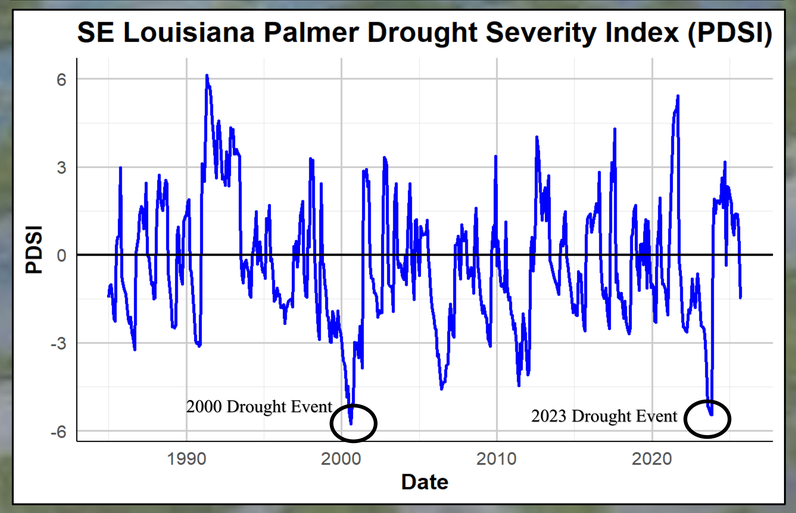
- Coastal wetlands are dynamic systems influenced by hydrological changes and climatic variability
- Vegetation browning outside of normal senescence indicates environmental stress
- The last drought event of this magnitude in 2000, led to the browning of over 100,000 acres of marsh as well as significant loss to coastal wetlands
- CRMS Network established in 2005, ~390 sites

Baseline NDVI **2023 Drought NDVI**



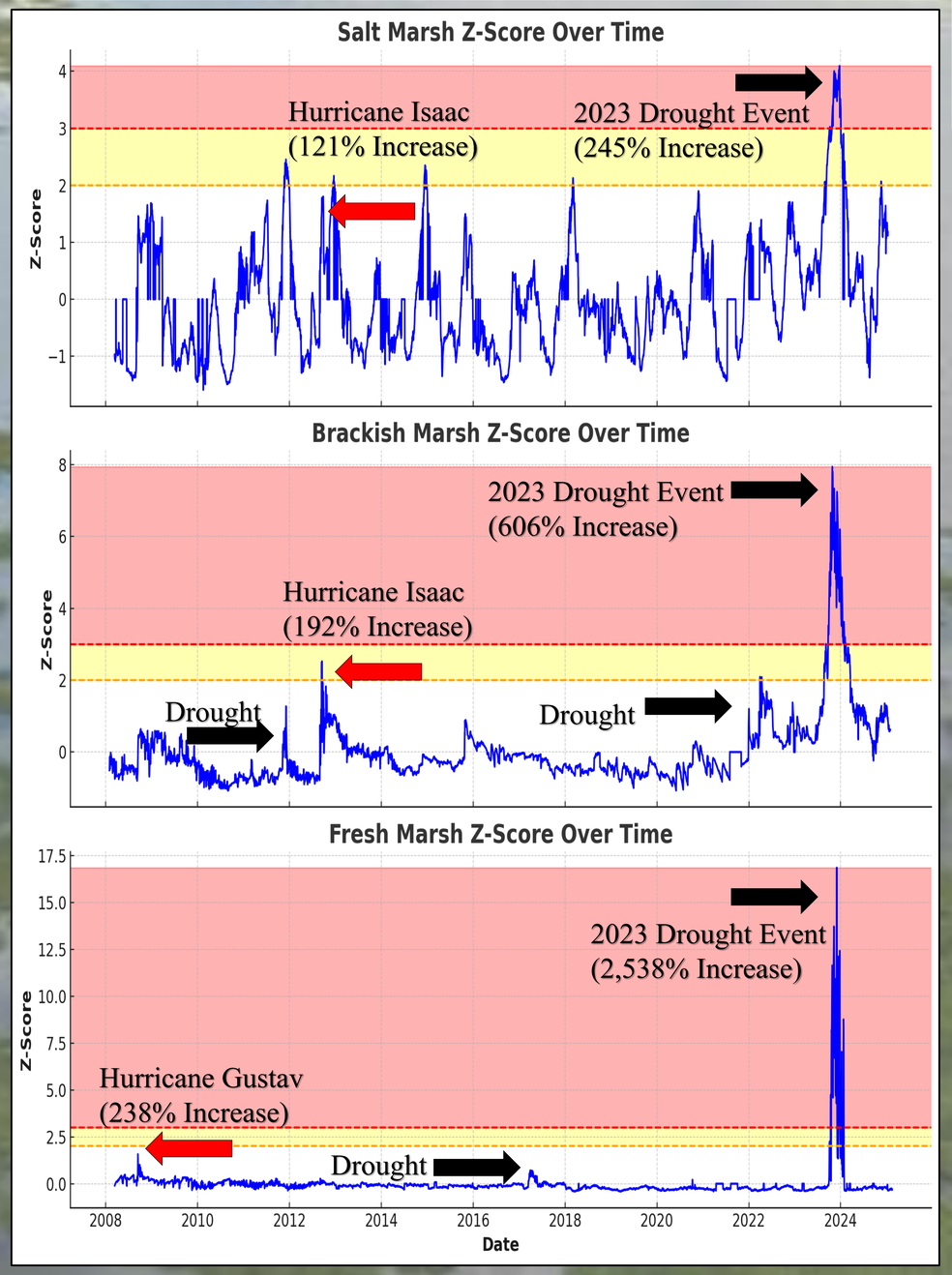
Research Questions

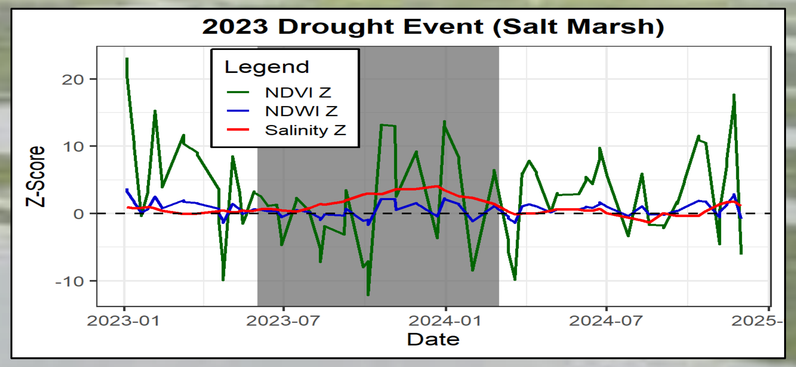
Q1: How do drought-driven salinity events differ temporally from storm surge salinity spikes?
Objective 1: Quantify drought vs hurricane salinity impacts
Q2: What is the relationship between salinity fluctuations and NDWI values over time?
Objective 2: Correlate vegetation indices with salinity data

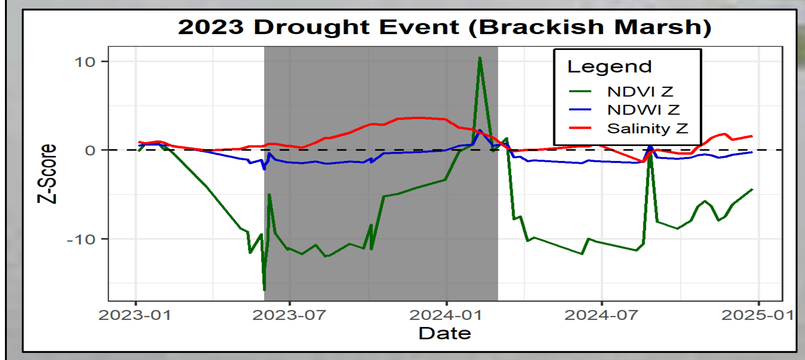


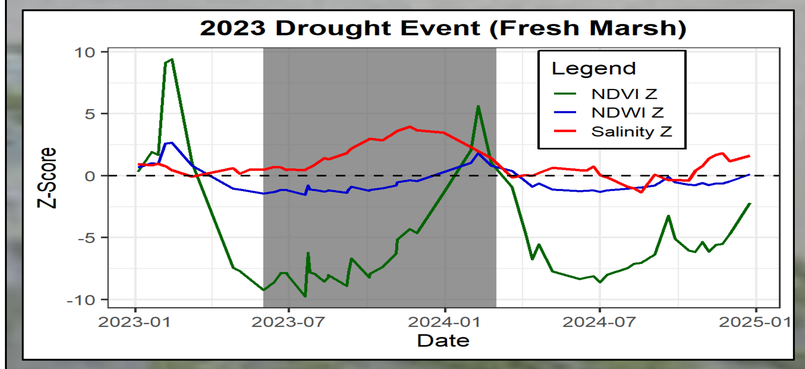
Results & Discussion

- The 2023 drought produced the strongest and most persistent salinity anomaly and caused multi-month elevation of salinity in each marsh type
- NDVI and NDWI declined during the drought, reflecting vegetation stress
- Outside drought periods, salinity–vegetation correlations were weak, indicating that vegetation becomes sensitive only once salinity exceeds seasonal norms.









Conclusions

- Drought-driven salinity intrusion produced longer-lasting and more extreme anomalies than storm surges, posing a greater sustained threat to marsh health.
- During normal years, vegetation indices fluctuate seasonally but remain centered around zero Z-score. During the 2023 drought NDVI had consistent anomalies.
- Brackish marshes were the most sensitive, showing the strongest coupling between salinity spikes and vegetation decline.

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